Soil & Feedstock Characterization Database

Executive Summary

Our analytical database contains complete geochemical characterization of both the deployment soils and the basalt feedstock used across our field trials and controlled experiments. This data enables precise site-matching: given a new deployment location's soil profile, we can predict the likely weathering trajectory and agronomic response before a single tonne of rock is applied.

What We Have

1. Baseline Soil Geochemistry (XRF Elemental Analysis)

Complete X-Ray Fluorescence analysis of deployment soils, providing oxide-level elemental composition:

Element	Unit	Sample 1	Sample 2
SiO_2	%	79.54	81.44
Al_2O_3	%	7.48	7.27
Fe_2O_3	%	2.04	2.02
MnO	%	0.11	0.112
MgO	%	0.41	0.37
CaO	%	0.62	0.51
Na_2O	%	0.49	0.49
K_2O	%	3.99	3.96
TiO_2	%	0.24	0.23
P_2O_5	%	0.21	0.20
LOI	%	3.9	3.39

2. Cation Exchange Capacity (CEC)

CEC determines the soil's ability to hold and exchange nutrient cations — a critical parameter for predicting ERW effectiveness:

Sample	CEC (meq/100g)
Site 1	6.4
Site 2	7.3

Lower CEC soils (< 10 meq/100g) are generally more responsive to basalt amendment, as the introduced cations (Ca^{2+}, Mg^{2+}) fill previously unoccupied exchange sites.

3. Total Organic & Inorganic Carbon (TOC/TIC)

Baseline carbon content is essential for distinguishing ERW-derived carbon from pre-existing soil carbon:

Sample	TOC (%)	TIC (%)
Site 1	1.08	0.15
Site 2	0.97	0.14

4. Particle Size Distribution (Soil)

Full laser diffraction particle size distribution data from 0.55 µm to 763 µm, enabling precise soil texture classification and hydraulic conductivity modeling.

5. Trace Metal Analysis

Complete heavy metal screening ensures environmental safety compliance:

Element	Unit	Detection Limit	Measured
Cr	ppm	68	68
Co	ppm	37	<37
Cu	ppm	40	<40
Ni	ppm	24	<24
V	ppm	17	17–22

All heavy metals are at or below detection limits, confirming environmental safety of the deployment soil.

Feedstock Characterization

Basalt Grain Size Distribution

Parameter	Value
Mean	152.7 µm
Standard Deviation	188.5 µm
Mode	401.2 µm
d10	3.9 µm
d50	38.5 µm
d90	447.5 µm

Specific Surface Area (BET)

Parameter	Value
Sample mass	1.61 g
Correlation coefficient	0.999973
BET area	2.3 m^2/g

The BET surface area is a critical predictor of initial weathering rate — higher surface area means faster initial dissolution but also faster passivation layer formation.

How We Use This Data

When a potential partner provides us with their target deployment location's soil report (XRF, CEC, pH), we can:

1. **Match it** against our characterized soil database to find the closest analogue
2. **Predict** the likely weathering trajectory based on observed outcomes from similar soils
3. **Flag risks** — high-clay soils with low hydraulic conductivity are flagged for crusting risk
4. **Optimize** the application rate and grain size to maximize Year 1 carbon capture

What We Are Not Sharing Here

- The exact predictive algorithms linking soil parameters to weathering outcomes
- The proprietary soil-matching database spanning multiple geographies
- Our calibrated weathering rate constants for each soil type
- The specific risk thresholds that trigger deployment warnings